Justin Ho

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EDUCATION

De Anza College Expected Jun. 2027

Associate in Science for Transfer in Computer Science (Intended Data Science Transfer) Cupertino, CA

- **GPA**: 3.66 | Dean's List (all terms)
- **Transfer**: Applying to transfer into a four-year Data Science program for Fall 2027 (expected B.S. 2029)
- **Coursework**: Data Structures (C++), Python, R, Assembly Language, Differential Equations, Linear Algebra, Discrete Mathematics, Statistics, Foundations of Data Science

EXPERIENCE

Mr. Chairman and Sons Feb. 2026 – Present

Furniture Production & Repair Assistant San Jose, CA

- Restore chairs end to end: strip worn materials, drill installation holes, and weave replacement straps to produce secure, consistent results.
- Measure, cut, and shape straps to exact specifications, where small errors compromise fit and durability.
- Work directly with the company owner on daily production and customer deliveries, completing assigned work independently.

Second Harvest Feb. 2026 – Present

Volunteer San Jose, CA

- Distribute groceries to community members at recurring Second Harvest food distributions.
- Work alongside a team of volunteers to ensure efficient, respectful service for all recipients.

PROJECTS

Visceral: Global Health Anatomy Explorer | *Python, SQL, React, Three.js* | Live Demo Oct. 2026

- Built a reproducible data pipeline (pandas, DuckDB SQL, Pandera) that processes **15.5M** WHO Global Health Estimates records across **185 countries** (2000–2023), mapping 218 causes of death to 31 organ systems.
- Showed that age standardization reorders country rankings (heart disease: Spearman $\rho = 0.25$ between crude and age-standardized ranks; Japan falls from 43rd to 180th), and quantified a **15%** bias from coarse age bins plus sensitivity to 5 standard populations.
- Deployed an interactive 3D web app (React, TypeScript, Three.js) where users click organs to explore burden by country, sex, age, and year; **180+** automated test cases run in CI via GitHub Actions.

Skillmap: Global Temperature Forecasting | *Python, scikit-learn, Streamlit* | Live Demo Oct. 2026

- Built an ML study forecasting next-day high temperature at **200 sites** across 29 Köppen climate classes (11 years of ERA5 data); gradient boosting beat the better of persistence and climatology at **191/200** sites (median **10.7%** lower error) on held-out years.
- Designed leakage-safe chronological evaluation in scikit-learn (year-based splits, fold-fitted climatology, paired block-bootstrap CIs), showing ML gains shrink where temperature anomalies persist (Spearman $\rho = -0.63$).
- Wrote a concurrent, resumable downloader for 2,200 site-years under API rate limits (shared multi-window limiter, global back-off on HTTP 429), reaching a **7.1x speedup** with 8 threads in a 48-request benchmark; deployed results as an interactive Streamlit map.

SKILLS

Languages : Python, C++, R, SQL, TypeScript, x86 Assembly

Tools : Git/GitHub, Jupyter Notebook, VS Code, NumPy, pandas, scikit-learn, Matplotlib, DuckDB, RStudio, Streamlit, Plotly, React, Three.js